

CLASSIFICATION

PERFORMANCE EVALUATION – PART I

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PERFORMANCE EVALUATION

The following concepts will be introduced:

- ✓ **TRAINING ERROR**
- ✓ **GENERALIZATION ERROR**
- ✓ **OVERFITTING/UNDERFITTING**

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	n	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	y	7	1.78	OH	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	n	6	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA	510	355-9993
147	0	157	n	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173



You know that a **POINT ESTIMATE OF THE CLASSIFICATION MODEL'S ACCURACY IS NOT ENOUGH** to be confident that the Classification Model you developed will provide a reliable prediction when queried on unseen records (new customers).

Furthermore, you are aware of the risk of **OVERFITTING/UNDERFITTING THE CUSTOMERS CALLS DATABASE** provided by the **CEO OF THE TELECOM COMPANY**.

You know that the **ERRORS COMMITTED BY A CLASSIFICATION MODEL** are divided into two types:

- ✓ **TRAINING ERROR**: the number of records of the **TRAINING SET** which are misclassified
- ✓ **GENERALIZATION ERROR**: expected error on previously unseen records (**TEST SET**)

A good Classification Model must not only fit the Training Set well, it must also accurately classify records it has never seen before (Test Set).

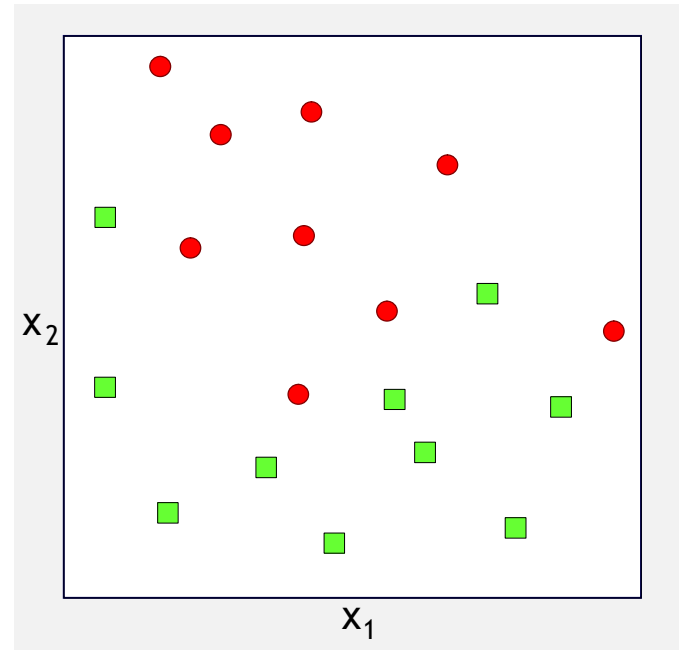
A good Classification Model must have low

- ✓ **TRAINING ERROR**
- ✓ **GENERALIZATION ERROR**

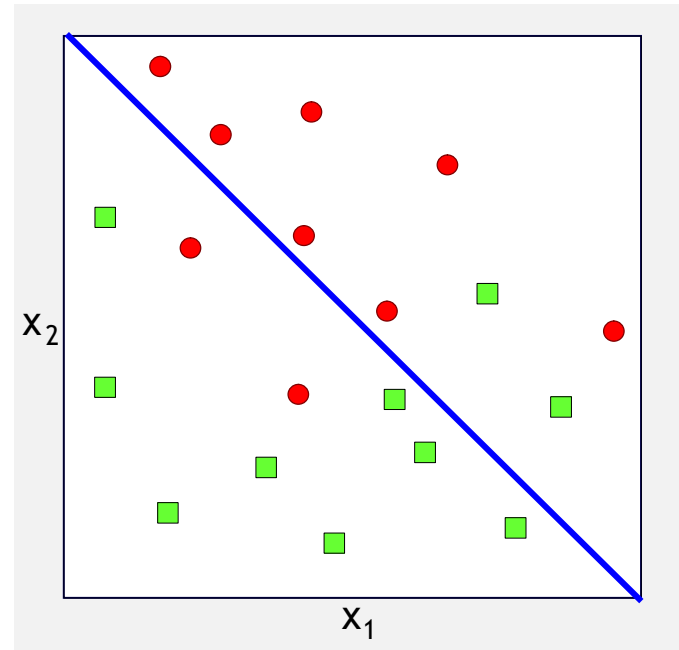
A **Classification Model fitting the Training Set too well** can have a **poorer Generalization Error** (estimated through the Test Set) than a model with higher Training Error.

MODEL OVERFITTING

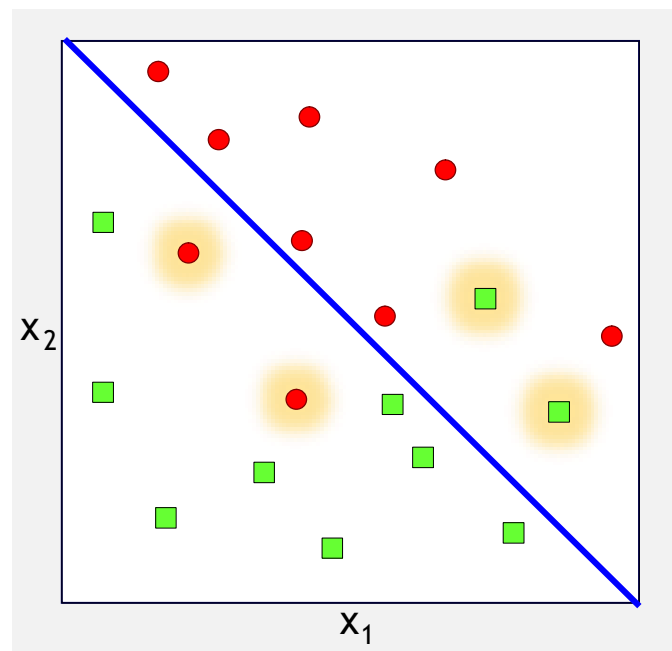
Consider a Training Set where **two NUMERIC INPUT ATTRIBUTES** are considered and a **BINARY CLASS ATTRIBUTE** must be predicted.



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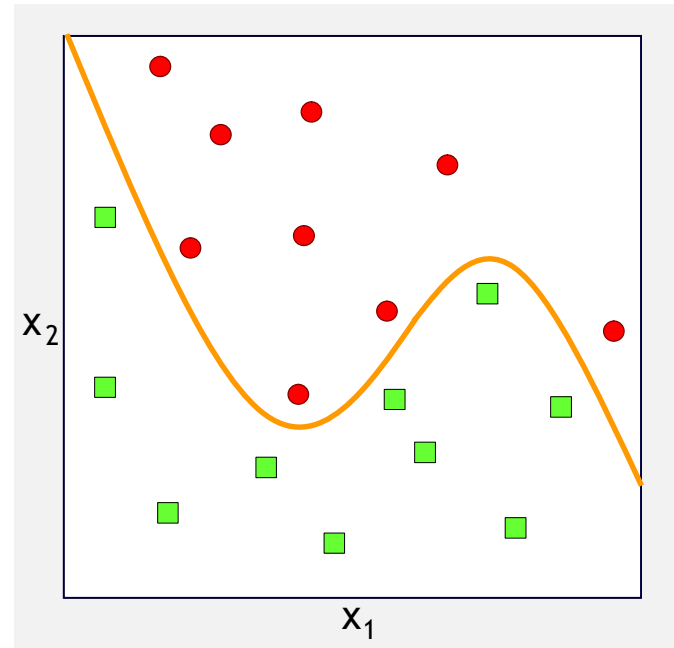


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4 records are wrongly labelled according to the blue line classification model (**Training Error**).

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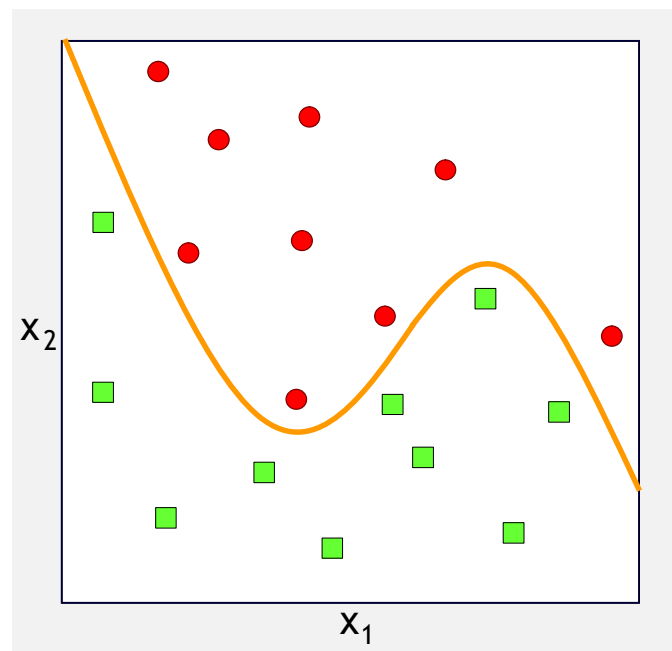


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Consider a Training Set where two **NUMERIC INPUT ATTRIBUTES** are considered and a **BINARY CLASS ATTRIBUTE** must be predicted.

Orange classification model achieves a better **Training Error** than the Blue Classification Model.

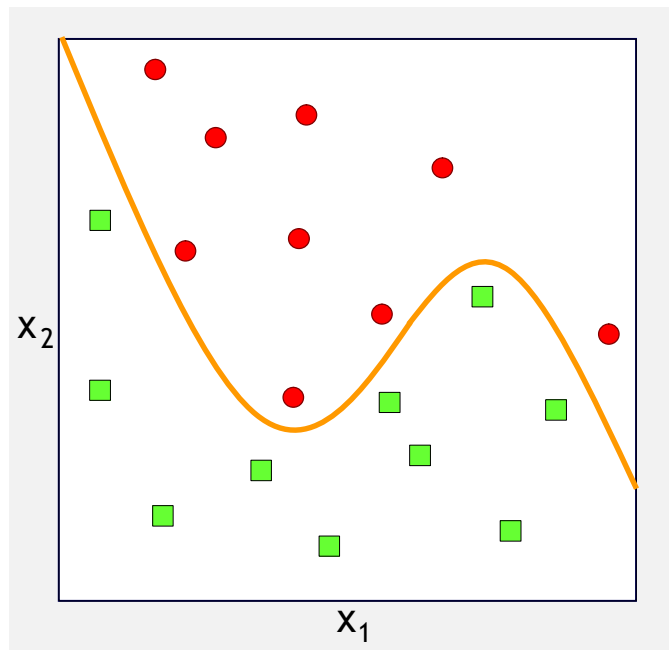


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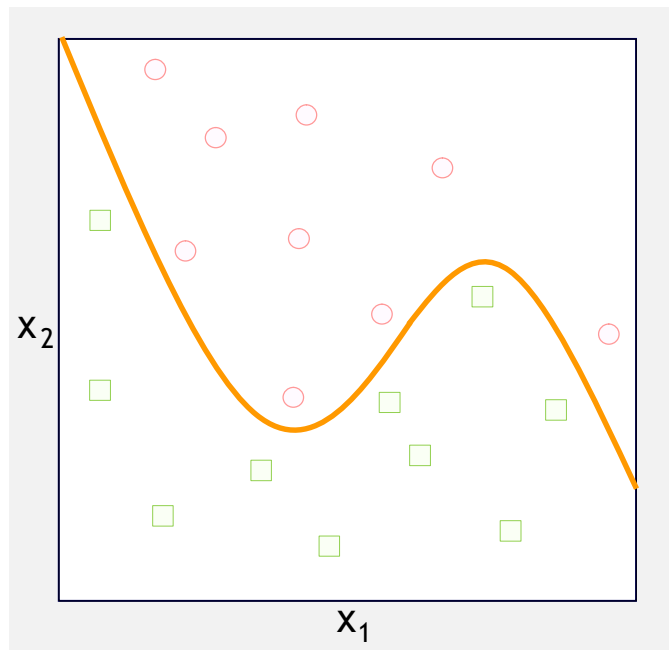
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What happens on the previously unseen records?

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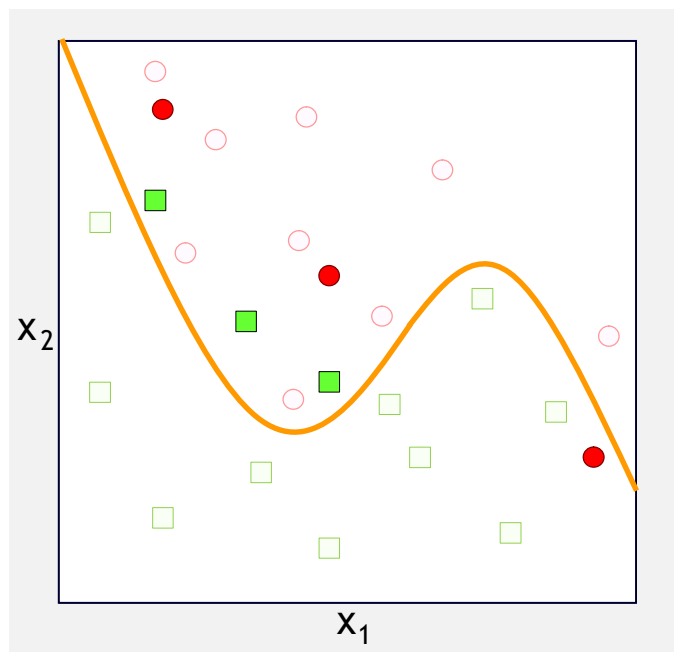
0 records are wrongly labelled according to the orange line classification model (**Training Error**).

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New records are previously unseen data (Test Set).



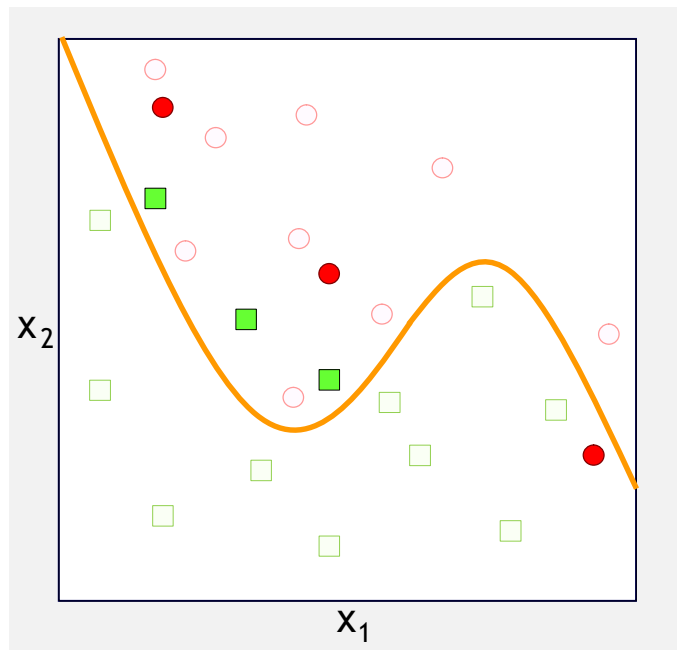
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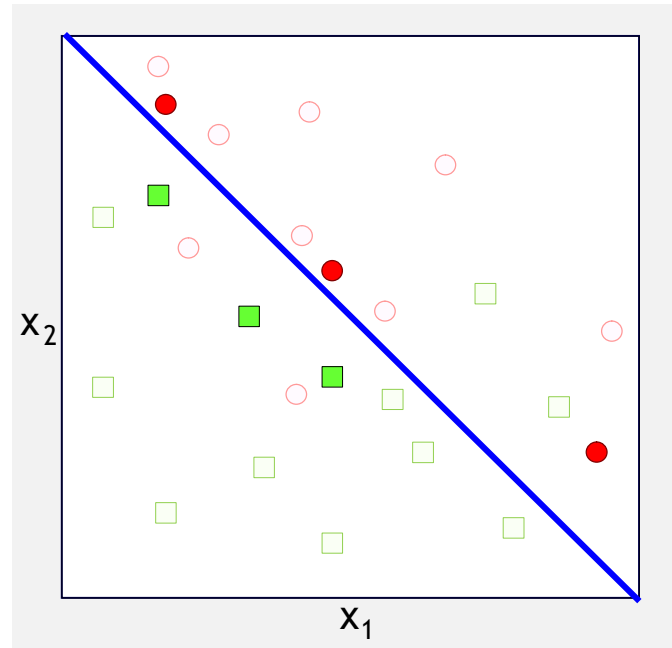
0 records are wrongly labelled according to the orange line classification model (**Training Error**).

4 records are wrongly labelled according to the orange line classification model (it estimates the **Generalization Error**).

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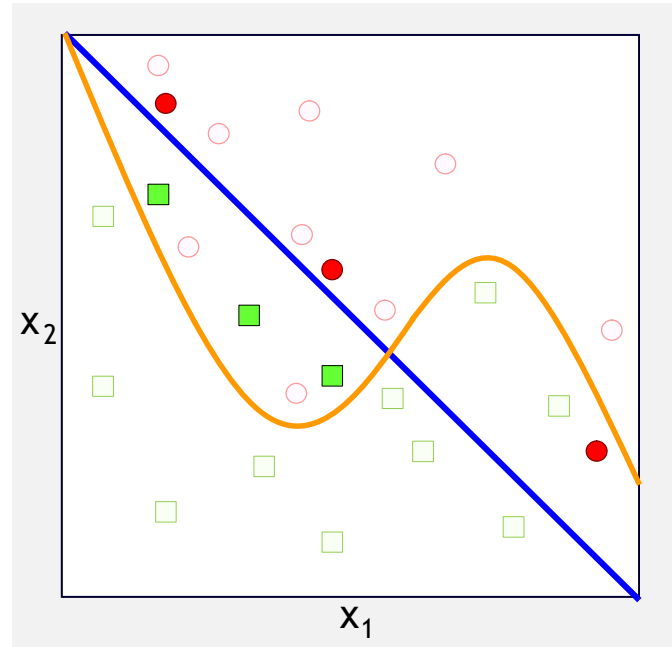
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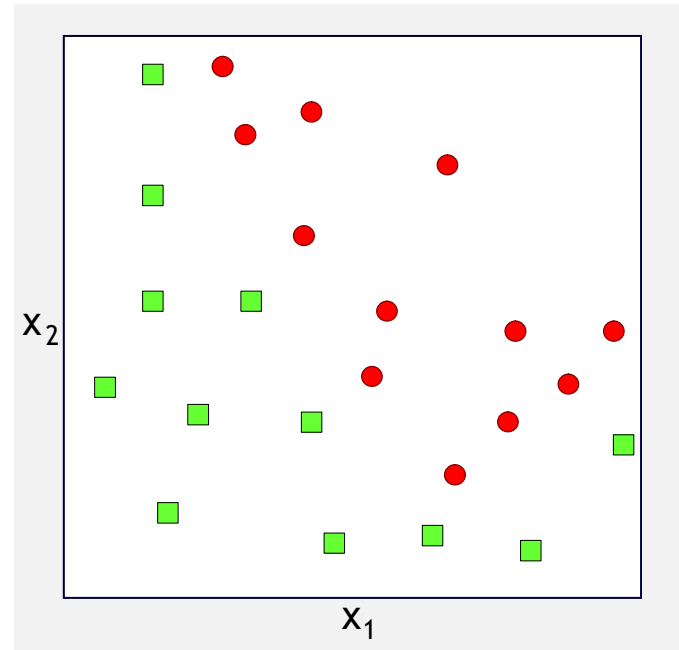
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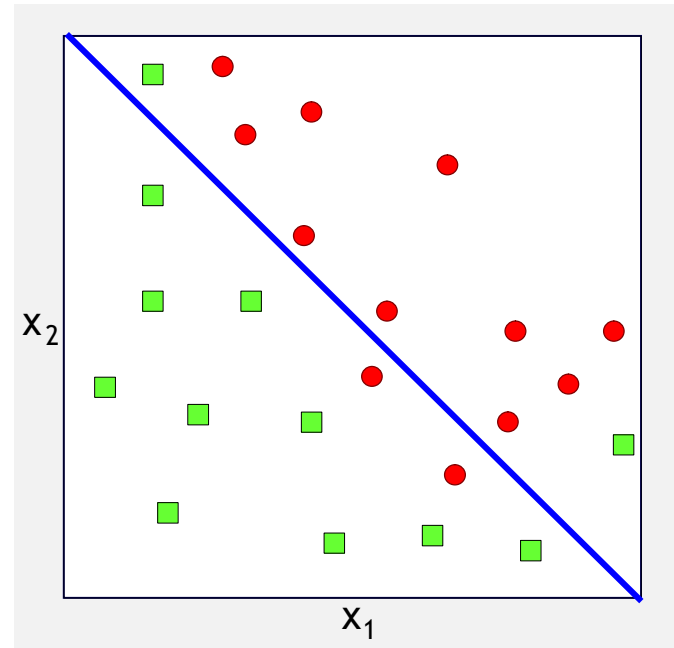
The Classification Model (**orange line**) achieving the **lowest** value of the **Training Error**, achieves the **highest** value of the **Generalization Error**, while the Classification Model (**blue line**) achieving the **highest** value of the **Training Error**, achieves the **lowest** value of the **Generalization Error**.

MODEL OVERFITTING

Also the opposite behavior can emerge.

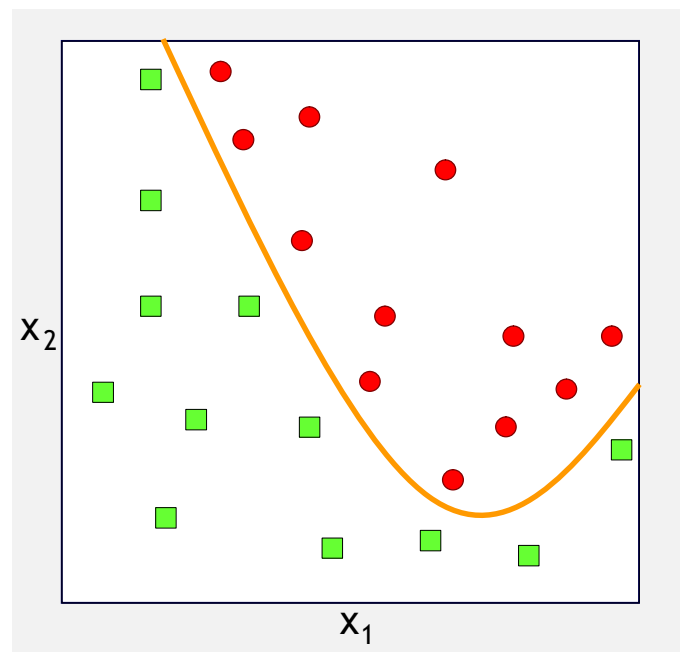


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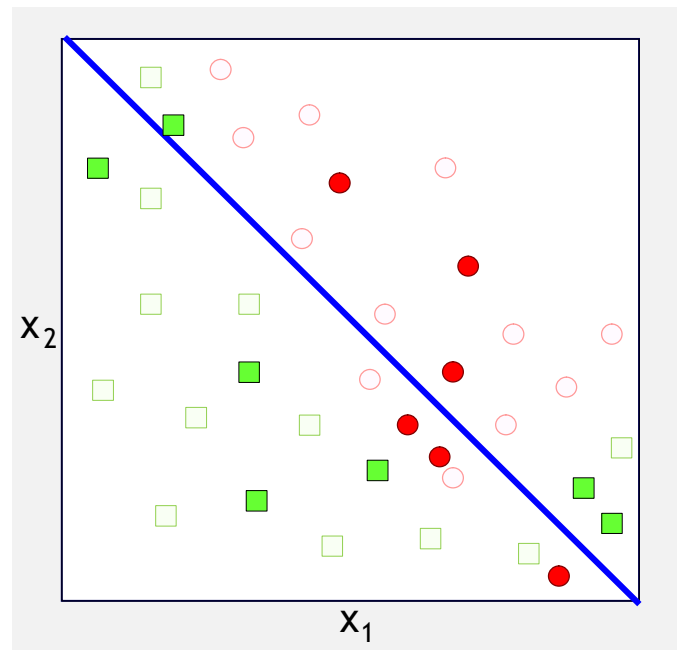


4 records are wrongly labelled according to the blue line classification model (**Training Error**).

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Also the opposite behavior can emerge.

6 records are wrongly labelled according to the blue line classification model (**Generalization Error**).



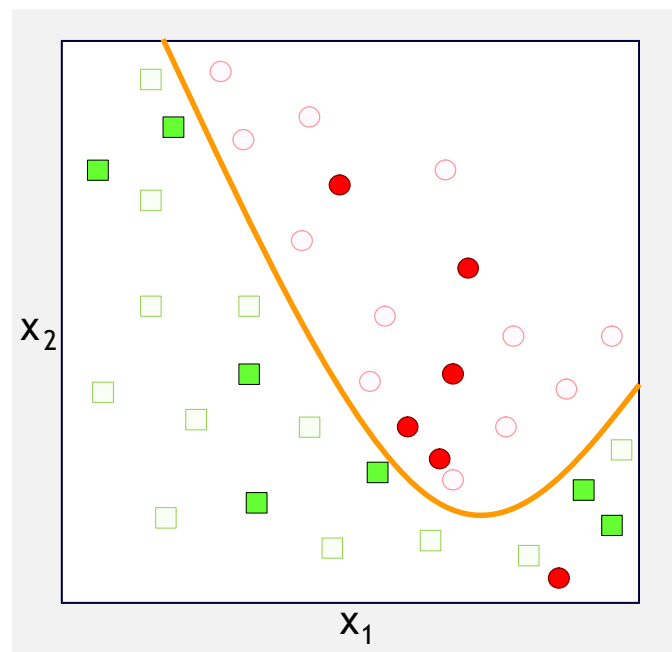
4 records are wrongly labelled according to the blue line classification model (**Training Error**).

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Also the opposite behavior can emerge.

6 records are wrongly labelled according to the blue line classification model (**Generalization Error**).

1 record is wrongly labelled according to the orange line classification model (**Generalization Error**).



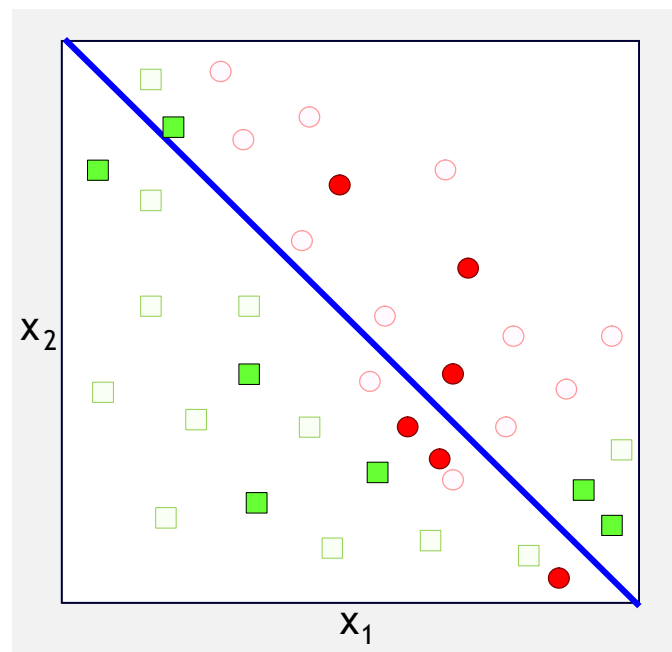
4 records are wrongly labelled according to the blue line classification model (**Training Error**).

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Also the opposite behavior can emerge.

6 records are wrongly labelled according to the blue line classification model (**Generalization Error**).

1 record is wrongly labelled according to the orange line classification model (**Generalization Error**).



4 records are wrongly labelled according to the blue line classification model (**Training Error**).

0 records are wrongly labelled according to the orange line classification model (**Training Error**).

Training and Test Error achieved by the blue line Classification Model are high on the Training Set as well as on the Test Set (4 on the Training Set and 6 on the Test Set).

MODEL UNDERFITTING